King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 102 Major Exam II Term 233

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. $7\int_0^{\pi}\sin(3x)\cos(4x)\,dx =$

(a) -4

(b) -6

(c) 0

(d) 6

(e) 4

2. $\int \tan^3(3t)\sec^3(3t)\,dt =$

(a) $\dfrac{\sec^3(3t)}{9}-\dfrac{\sec(3t)}{3}+C$

(b) $\dfrac{\sec^4(3t)}{12}-\dfrac{\sec^2(3t)}{6}+C$

(c) $\dfrac{\tan^3(3t)}{9}-\dfrac{\tan(3t)}{3}+C$

(d) $\dfrac{\tan(3t)}{15}-\dfrac{\tan^3(3t)}{9}+C$

(e) $\dfrac{\sec^5(3t)}{15}-\dfrac{\sec^3(3t)}{9}+C$

3. $\int_0^{\pi/3} x\sec^2 x\,dx =$

(a) $\dfrac{\pi}{2}-\ln 3$

(b) $\dfrac{\sqrt{3}}{3}\pi+\ln 2$

(c) $\dfrac{\pi}{2}+\ln 3$

(d) $\dfrac{\sqrt{3}}{3}\pi-\ln 2$

(e) $\dfrac{\sqrt{3}}{3}\pi-\ln 3$

4. $\int \dfrac{\sqrt{x^2+1}}{x^4}\,dx =$

(a) $-(x^2+1)^{3/2} + \dfrac{1}{3x^3} + c$

(b) $-\dfrac{(x^2+1)^{3/2}}{3x^3} + c$

(c) $-\dfrac{x^2}{\sqrt{x^2+1}} - 3(x^2+1)^3 + c$

(d) $-\dfrac{1}{3(x^2+1)^{3/2}} - \dfrac{(x^2+1)^{3/2}}{x^3} + c$

(e) $-\dfrac{3x^3}{\sqrt{x^2+1}} + c$

5. The improper integral $\int_2^\infty \dfrac{1}{\sqrt{x}-1}\,dx$

(a) converges to 1

(b) converges to 3

(c) converges to $\dfrac{3}{2}$

(d) converges to 2

(e) diverges

6. $\int_{\pi/3}^{\pi/2} \dfrac{dx}{\cos x - 1} =$

(a) $\sqrt{3}+1$

(b) $1-\sqrt{3}$

(c) $\sqrt{3}-1$

(d) $1-\sqrt{2}$

(e) $\sqrt{2}+1$

7. The volume of the solid formed by revolving the region in the first quadrant bounded by the graphs of $y = 3x - x^2$, $y = 0$ about the line $x = 4$ is

(a) 24π

(b) 21π

(c) 22π

(d) $\dfrac{45\pi}{2}$

(e) $\dfrac{35\pi}{2}$

8. The volume of the solid formed by revolving the region bounded by the graphs of $y = x^2 + 2$, $y = 0$, $x = 0$ and $x = 2$ about the y-axis is

(a) 2π

(b) 16π

(c) 4π

(d) π

(e) 8π

9. The improper integral $\int_0^\infty xe^{-2x}\,dx$

(a) converges to 2

(b) converges to $\dfrac{1}{4}$

(c) converges to $\dfrac{1}{2}$

(d) converges to 0

(e) diverges

10. $\int_0^{\sqrt{3}/2} \dfrac{1}{(1-x^2)^{5/2}}\,dx =$

(a) $4\sqrt{3}$

(b) $6\sqrt{3}$

(c) 3

(d) $3\sqrt{3}$

(e) $2\sqrt{3}$

11. The arc length of the graph of $x = \dfrac{1}{3}\left(y^2+2\right)^{3/2}$, $0 \le y \le 2$ is equal to

(a) $\dfrac{16}{3}$

(b) $\dfrac{3}{4}$

(c) $\dfrac{13}{3}$

(d) $\dfrac{8}{3}$

(e) $\dfrac{14}{3}$

12. The area of the surface generated by revolving the graph of $y = \sqrt{25 - x^2}$, $0 \leq x \leq 4$ about the y-axis is equal to:

(a) 16π

(b) 10π

(c) 24π

(d) 30π

(e) 20π

13. $\int \dfrac{x^2 + x - 1}{x^3 + 2x^2 + x}\, dx =$

(a) $-\ln|x| - 2\ln|x+1| - \dfrac{1}{x+1} + C$

(b) $-2\ln|x| + \ln|x+1| - \dfrac{1}{x+1} + C$

(c) $\ln|x| - 2\ln|x+1| - \dfrac{1}{x+1} + C$

(d) $-\ln|x| + 2\ln|x+1| - \dfrac{1}{x+1} + C$

(e) $2\ln|x| - \ln|x+1| - \dfrac{1}{x+1} + C$

14. $\int_0^1 \dfrac{x^2 - x}{x^2 + x + 1}\, dx =$

(a) $\ln 3$

(b) $1 - \ln 3$

(c) $\ln 3 - 1$

(d) $1 + \ln 3$

(e) 1

Answer Key

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Q	Ans	Brief explanation
1	**(b)**	$\sin 3x \cos 4x = \frac{1}{2}(\sin 7x - \sin x)$, so $7\int = \frac{7}{2}\left(\frac{2}{7} - 2\right) = -6$.
2	**(e)**	$u = \sec 3t$: $\frac{1}{3}\int (u^4 - u^2)\, du = \frac{\sec^5 3t}{15} - \frac{\sec^3 3t}{9} + C$.
3	**(d)**	By parts: $[x \tan x]_0^{\pi/3} - \int_0^{\pi/3} \tan x\, dx = \frac{\sqrt{3}}{3}\pi - \ln 2$.
4	**(b)**	$x = \tan\theta$: $\int \frac{\cos\theta}{\sin^4\theta} d\theta = -\frac{1}{3\sin^3\theta} = -\frac{(x^2+1)^{3/2}}{3x^3} + c$.
5	**(e)**	$\frac{1}{\sqrt{x}-1} > \frac{1}{\sqrt{x}}$ and $\int_2^\infty x^{-1/2} dx$ diverges, so it diverges.
6	**(b)**	$u = \tan\frac{x}{2}$: integrand becomes $-u^{-2} du$, so $[1/u]_{1/\sqrt{3}}^1 = 1 - \sqrt{3}$.
7	**(d)**	Shells, radius $4 - x$: $2\pi \int_0^3 (4 - x)(3x - x^2)\, dx = \frac{45\pi}{2}$.
8	**(b)**	Shells: $V = 2\pi \int_0^2 x(x^2 + 2)\, dx = 2\pi(4 + 4) = 16\pi$.
9	**(b)**	By parts, $\lim_{b\to\infty} \left[-\frac{x}{2}e^{-2x} - \frac{1}{4}e^{-2x}\right]_0^b = \frac{1}{4}$.
10	**(e)**	$x = \sin\theta$: $\int_0^{\pi/3} \sec^4\theta\, d\theta = [\tan\theta + \frac{\tan^3\theta}{3}] = 2\sqrt{3}$.
11	**(e)**	$1 + (x')^2 = (y^2 + 1)^2$, so $L = \int_0^2 (y^2 + 1)\, dy = \frac{14}{3}$.
12	**(e)**	$S = 2\pi \int_0^4 \frac{5x}{\sqrt{25 - x^2}}\, dx = 10\pi(5 - 3) = 20\pi$.
13	**(d)**	$\frac{-1}{x} + \frac{2}{x+1} + \frac{1}{(x+1)^2}$, so $-\ln\lvert x\rvert + 2\ln\lvert x + 1\rvert - \frac{1}{x+1} + C$.
14	**(b)**	$= 1 - \frac{2x+1}{x^2+x+1}$, so the integral is $[x - \ln(x^2 + x + 1)]_0^1 = 1 - \ln 3$.